

WHATI, NT, Canada

WMO: 711630

Lat: 63.134N

Lon: 117.244W

Elev: 271

StdP: 98.11

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.9	-38.9	-45.1	0.0	-40.4	-43.0	0.1	-38.7	7.1	-17.9	6.4	-20.3	0.7	270	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	27.3	16.3	25.4	15.7	23.5	15.0	17.7	24.5	16.8	23.3	15.9	22.1	3.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.3	11.2	19.8	14.2	10.5	18.8	13.2	9.8	18.1	51.0	24.6	48.2	23.4	45.7	22.2	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.5	5.9	-43.0	30.2	3.1	1.9	-45.3	31.6	-47.1	32.7	-48.9	33.8	-51.1	35.1		
			-42.7	19.4	2.9	0.7	-44.8	19.9	-46.5	20.4	-48.1	20.7	-50.3	21.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.4	-25.1	-22.3	-17.3	-5.9	5.5	13.5	16.4	13.1	6.8	-2.4	-14.2	-22.4
	DBStd	16.07	7.92	7.14	8.30	7.39	5.86	4.39	3.57	3.81	4.31	4.64	6.89	8.30
	HDD10.0	5735	1088	905	847	476	161	17	2	13	114	384	725	1005
	HDD18.3	8341	1346	1138	1105	726	398	154	81	166	346	642	975	1263
	CDD10.0	470	0	0	0	0	21	121	199	111	18	0	0	0
	CDD18.3	33	0	0	0	0	0	8	20	5	0	0	0	0
	CDH23.3	444	0	0	0	0	13	120	226	82	3	0	0	0
	CDH26.7	81	0	0	0	0	1	24	42	14	1	0	0	0
Wind	WSAvg	2.8	2.0	2.3	2.7	3.0	3.3	3.3	3.1	3.1	3.2	2.9	2.3	1.7
Precipitation	PrecAvg	298	13	10	13	10	18	36	48	50	39	27	19	15
	PrecMax	354	25	20	27	16	35	93	95	82	79	42	37	27
	PrecMin	224	5	4	3	4	4	7	25	19	11	11	10	5
	PrecStd	35	5	4	6	4	9	18	17	17	15	8	6	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.0	-2.1	6.7	13.9	25.3	29.6	30.1	28.3	22.1	11.4	0.7	-2.3
		MCWB	-2.6	-3.3	2.5	6.9	12.8	16.4	17.8	17.0	14.1	8.4	0.2	-2.9
	2%	DB	-6.3	-6.9	2.5	10.5	21.5	26.6	27.7	25.6	18.8	7.6	-1.2	-5.7
		MCWB	-6.7	-7.5	-0.2	4.9	11.5	15.1	16.8	16.1	12.2	5.2	-1.7	-5.9
	5%	DB	-11.3	-9.4	-0.7	8.2	18.6	24.3	25.8	23.2	16.6	5.2	-3.2	-8.2
		MCWB	-11.6	-9.9	-2.4	3.5	10.1	14.4	16.3	15.5	11.4	4.0	-3.5	-8.5
	10%	DB	-14.5	-11.8	-3.9	5.5	15.4	22.1	24.0	21.0	14.2	3.5	-4.9	-10.8
		MCWB	-14.8	-12.2	-5.2	1.9	8.5	13.5	15.9	14.5	10.4	2.5	-5.2	-11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.5	-3.2	2.7	6.9	13.6	17.6	19.2	18.3	15.0	8.3	0.4	-3.0
		MCDB	-2.0	-1.9	6.2	13.3	22.7	26.9	25.5	25.3	19.8	10.9	1.0	-2.4
	2%	WB	-6.7	-7.5	0.0	5.1	11.8	16.3	18.1	17.2	13.4	5.7	-1.6	-6.0
		MCDB	-6.2	-6.9	2.1	10.2	20.3	24.1	24.5	23.4	17.3	7.1	-1.2	-5.7
	5%	WB	-11.6	-9.9	-2.5	3.5	10.3	15.3	17.3	16.1	12.0	3.9	-3.5	-8.4
		MCDB	-11.3	-9.5	-0.7	7.5	17.8	22.4	23.8	21.9	15.7	5.0	-3.2	-8.1
	10%	WB	-14.8	-12.2	-5.3	2.0	8.8	14.2	16.5	15.1	10.7	2.5	-5.2	-11.1
		MCDB	-14.5	-11.8	-4.0	5.1	15.1	20.7	22.8	19.9	14.0	3.4	-4.9	-10.8
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.5	13.9	13.9	11.7	11.3	11.3	10.8	9.1	5.9	7.7	8.3
		MCDBR	9.4	12.0	15.7	15.3	15.9	14.7	14.5	15.2	13.6	9.0	7.3	9.9
		MCWBR	9.3	11.3	12.9	10.3	8.0	6.3	6.3	7.5	7.9	6.7	7.0	9.7
	5% WB	MCDBR	9.2	11.9	15.2	14.2	15.1	13.0	12.4	12.7	12.1	8.1	6.9	9.8
		MCWBR	9.1	11.3	12.5	9.7	7.9	6.0	6.0	6.9	7.6	6.1	6.6	9.7
Clear-Sky Solar Irradiance	taub		0.180	0.228	0.257	0.298	0.327	0.338	0.369	0.335	0.304	0.262	0.177	0.149
	taud		2.017	2.262	2.268	2.254	2.445	2.467	2.394	2.528	2.569	2.453	2.035	1.943
	Ebn at Noon		493	754	861	887	890	885	840	846	811	708	485	315
	Edh at Noon		29	55	83	106	97	97	102	82	64	47	27	14
All-Sky Solar Radiation	RadAvg		0.26	1.09	2.91	5.01	5.84	6.08	5.30	4.00	2.40	0.89	0.28	0.11
	RadStd		0.02	0.06	0.14	0.26	0.39	0.59	0.28	0.27	0.23	0.11	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	+1.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page